Implementing Zero Trust Network Access and Segmentation for Enterprise VR Systems

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D833 Cybersecurity Information and Assurance Capstone

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Table of Contents

[Title]..... 3
Problem..... 3
Scope..... 3
Stakeholders..... 3
Risk Assessment..... 3
Objectives..... 3
Proposed Cybersecurity Technical Control Solution..... 3
Industry Standards and Compliance Regulations..... 3
Integration..... 3
Audit Trail and Reporting..... 3
Justification..... 3
Cost..... 3
Implementation Plan..... 4
Resources..... 4
Dependencies..... 4
Risk Mitigation..... 4
Benefits and Impact..... 4
Security Benefits..... 4
Operational Impact..... 4
Business Impact..... 4
Conclusion..... 4
References..... 5
Appendix A..... 6

Implementing Zero Trust Network Access and Segmentation for Enterprise VR Systems

For my capstone project, I am focusing on a fictional organization named VirtuTrain Solutions. VirtuTrain is a privately held software development company with 450 employees operating in the corporate education sector, providing custom VR training simulations to large enterprise clients. To facilitate immersive training and team collaboration, they are rolling out 150 Meta Quest Pro headsets to their distributed developers and quality assurance testers. The existing technology infrastructure relies on a standard Cisco AnyConnect VPN for access, where employee laptops utilize CrowdStrike endpoint protection and Active Directory authentication.

However, the newly deployed VR headsets lack these endpoint security agents. To function properly, this hardware must continuously collect and process a massive amount of sensitive telemetry data, including mapping the user's physical surroundings, recording audio, and tracking biometrics like eye and facial movements. Introducing these IoT systems into an enterprise network presents unique challenges. According to NIST, organizations deploying IoT hardware must account for the fact that they often lack the built in security capabilities of traditional IT equipment, making them inherently difficult to secure and monitor (Boyson et al., 2018). The primary assets that need protection in this scenario are the mission critical Linux source code servers and the strict confidentiality of the employees' biometric data.

Problem

The primary cybersecurity problem is the severe risk created by integrating unmanaged VR headsets into the existing remote access infrastructure. Remote employees are connecting these devices to the internal corporate environment using the standard VPN without any endpoint verification. This setup essentially bridges poorly secured home Wi-Fi connections straight into

the primary IT infrastructure, placing the headsets on the exact same segment as mission critical source code servers.

As a result of this flat topology, the hardware transmits highly sensitive, unencrypted biometric and spatial data without any isolation. This creates a massive threat to the confidentiality of employee personal information. Because these remote endpoints lack standard enterprise protections or posture checks, they represent a critical control weakness.

Telework environments require strict access controls, and allowing unverified hardware to bypass these protections dramatically increases the attack surface (Souppaya & Scarfone, 2016). In this case, the attack vector is clear. If a threat actor compromises a user's home router or the headset itself, they could easily use that trusted VPN tunnel to pivot into the main enterprise environment. From there, an attacker could access confidential servers or deploy ransomware. This would severely impact the organization's overall system confidentiality, integrity, and availability, requiring immediate remediation to prevent a catastrophic data breach.

Scope

The scope of this project focuses specifically on the 150 organizationally issued Meta Quest Pro VR headsets, the central Cisco AnyConnect VPN gateway, and the internal routing rules needed to create a new /24 VLAN subnet. It also includes securing all spatial and biometric data generated during remote sessions.

This project explicitly excludes the 450 standard employee Windows 11 laptops and the two primary Linux source code servers. While those systems will directly benefit from the added network isolation, their existing Active Directory authentication and CrowdStrike endpoint protection serve as adequate compensating controls.

Stakeholders

Successfully implementing this Zero Trust architecture requires coordination across three key groups to ensure a secure and efficient rollout. The IT Network Engineering Team, the Information Security Team, and Executive Management will each play distinct roles in bridging the gap between technical execution and business risk management.

The IT Network Engineering Team serves as the subject matter experts for the organization's routing infrastructure. They will design and deploy the dedicated VLAN subnet while configuring the new access policies on the VPN gateway. This technical execution is critical for physically separating the traffic and protecting the core environment.

The Information Security Team will define the specific endpoint posture checks within the MDM solution and establish the encryption standards for the remote headsets. Their expertise guarantees that the final deployment complies with industry privacy regulations and adequately mitigates the identified vulnerabilities.

Finally, Executive Management serves as the ultimate risk owners for the organization. They will provide necessary project oversight, review the proposed mitigation strategies, approve the budget, and authorize the final rollout across the distributed workforce.

Risk Assessment

A primary concern stemming from the current VPN configuration is the risk of lateral movement and potential ransomware infection. Unmanaged VR headsets connected via the standard VPN could be compromised and used as a gateway to access the internal corporate network. This threat would likely originate from external cybercriminals or state sponsored actors deploying malware or exploiting unpatched vulnerabilities on the employee home network

or the VR headset itself. The core vulnerability lies in the lack of endpoint posture checks and the absence of network segmentation for VPN connections, which directly contradicts established NIST secure telework guidelines (Souppaya & Scarfone, 2016). If exploited, the potential impact includes unauthorized access to mission critical Linux source code servers, leading to intellectual property theft or widespread ransomware deployment. I assess the likelihood of this risk as high because employee home networks typically lack enterprise grade security, and unmanaged IoT devices are frequent targets for automated exploitation scripts. The severity is equally high. A successful attack of this nature would completely halt business operations and result in catastrophic financial and reputational damage.

Beyond the structural network vulnerabilities, the organization also faces a critical risk regarding biometric data exposure and privacy breaches. Highly sensitive biometric and spatial telemetry data transmitted by the VR headsets could easily be intercepted during transit. The primary threat actors in this scenario are malicious individuals performing man in the middle attacks on unsecured public Wi-Fi or compromised home routers. This vulnerability exists because the sensitive telemetry data is currently being transmitted without mandated strong encryption protocols like IPsec or TLS. The impact of such an interception would be the exposure of highly personal employee data, leading to severe privacy violations and regulatory compliance penalties, as biometric telemetry requires strict safeguarding measures (XR Safety Initiative, 2020). I consider the likelihood of this occurring to be medium. While intercepting specific network traffic requires a targeted effort, the continuous stream of unencrypted telemetry data provides ample opportunity for capture. However, the severity of this risk remains high. According to the organizational risk framing guidelines in NIST SP 800-39, events that

completely halt mission critical business operations and result in catastrophic financial and reputational damage must be prioritized at the highest tier (Joint Task Force Transformation Initiative, 2011). If a threat actor successfully intercepts and leaks biometric profiles, it would trigger this exact level of systemic failure due to massive compliance penalties and a complete loss of enterprise client trust.

Literature Support and Analysis

To conduct this evaluation, the methodology relied heavily on the guidelines provided in NIST SP 800-30 Revision 1 (Joint Task Force Transformation Initiative, 2012) and NIST SP 800-39 (Joint Task Force Transformation Initiative, 2011). While NIST SP 800-30 outlines a structured approach for identifying threat sources and mapping them to specific vulnerabilities to determine overall risk levels, NIST SP 800-39 provides the foundation for framing these risks within the broader organizational risk context.

These literature sources directly informed the risk assessment approach by providing a standardized framework to evaluate both the likelihood of a threat event occurring and the severity of its potential business impact. By applying this specific methodology, the assessment successfully isolated the lateral movement and data exposure risks as critical priorities. This structured analysis confirms that the organizational consequences of leaving these vulnerabilities unaddressed would be severe and require immediate remediation.

Objectives

The primary objective of this proposal is to establish a secure telework environment for all remote VR users. Achieving this overarching goal requires three targeted technical

milestones. First, IT will implement strict network segmentation by creating a dedicated VLAN to isolate all VR traffic away from the mission critical source code servers. Furthermore, the organization will enforce automated endpoint posture checks using an MDM solution to ensure no unmanaged device can connect to the corporate network. Finally, the proposed solution mandates strong encryption protocols for all spatial and biometric telemetry data while it is in transit.

Proposed Cybersecurity Technical Control Solution

The proposed solution utilizes a Zero Trust architecture to secure the VR headsets and protect the broader corporate network. The core of this initiative involves deploying an MDM platform to enforce strict endpoint posture checks. Before any headset is permitted to connect via the Cisco VPN, the MDM will verify that the hardware is running approved firmware and has not been compromised.

Simultaneously, the IT Network Engineering Team will configure a dedicated /24 VLAN specifically for VR traffic, segmenting the headsets entirely away from the main corporate infrastructure. Finally, all telemetry data transmitted by the endpoints will be encrypted using TLS 1.3. This multi-layered technical control directly mitigates the risks of lateral network movement and biometric data interception.

Industry Standards and Compliance Regulations

To ensure the proposed solution meets rigorous security requirements, it aligns directly with prominent industry frameworks. The implementation of endpoint posture checks and network segmentation adheres to the Zero Trust principles outlined in NIST SP 800-207 (Rose et al., 2020), which mandates continuous verification before granting network access. Furthermore,

because the VR headsets collect highly sensitive physical and behavioral data, the encryption and isolation controls comply with the strict privacy mandates established by the XRSI Privacy Framework (XR Safety Initiative, 2020). This standard specifically governs the ethical handling and secure transmission of immersive tracking data.

Integration

This technical solution is designed to seamlessly integrate with VirtuTrain's existing infrastructure. The new VLAN will be configured directly on the established Cisco routing hardware, requiring no new physical equipment. Additionally, the MDM posture checks will interface with the Cisco AnyConnect VPN gateway by utilizing API calls to verify device health before establishing the tunnel. The impact on legacy systems is intentionally minimal. Because standard employee laptops and Linux servers are excluded from this specific VLAN, their daily operations and CrowdStrike endpoint protections will remain completely unaffected.

Audit Trail and Reporting

The proposed solution meets strict compliance auditing and logging requirements by utilizing centralized data collection. The Cisco AnyConnect VPN gateway will be configured to log all connection attempts, including successful authorizations and blocked connections resulting from failed MDM posture checks. These logs will automatically forward to the organization's existing central logging server for retention and continuous monitoring. Additionally, the MDM platform will generate automated weekly compliance reports detailing the firmware versions and security status of all 150 VR headsets. This ensures that administrators have a clear and auditable trail of all remote VR access activities.

Justification

This solution is highly justified when compared to alternative approaches. One alternative considered was issuing a second, physically separate corporate laptop to every VR user to act as a dedicated tethered bridge. However, this approach is prohibitively expensive and creates a poor user experience. Another alternative was relying solely on VPN split tunneling to route VR traffic to the public internet. While cost-effective, split tunneling fails to provide any endpoint verification or protect the biometric data from local network interception. The proposed MDM and VLAN solution represents a superior strategy because it actively verifies device health and physically isolates the traffic. This approach offers a true Zero Trust environment and robust data protection while remaining highly efficient and cost-effective.

Cost

The estimated budget for this project includes all necessary hardware, software, licensing, personnel, and ongoing maintenance. Because the solution utilizes the existing Cisco networking infrastructure, physical equipment expenses are kept to an absolute minimum and are restricted to expanding the storage capacity of the internal logging server. The primary financial requirements involve the software licenses for the new MDM platform and the specialized VPN posture checking modules. Labor calculations are based on the estimated hours for the internal IT and security teams during the six-week deployment phase, while annual vendor support agreements cover continuous operational maintenance.

Resource Category	Description	Estimated Cost
Hardware	Dedicated solid state storage drives for the centralized logging server	$1,200

Software and Licensing	MDM Enterprise cloud licenses for 150 remote VR devices	$15,000
Software and Licensing	Cisco AnyConnect VPN Apex license upgrade for posture checking	$5,000
Personnel	IT Network Engineering labor for VLAN routing configuration	$3,000
Personnel	Information Security labor for posture check design and testing	$3,400
Maintenance	Annual vendor support and maintenance contract for the MDM platform	$2,500
Total Estimated Cost	**Project Total**	**$30,100**

Implementation Plan

The project rollout will occur over a six-week timeline divided into four distinct phases. During the initial design phase in weeks one and two, the Information Security Team will define the necessary MDM posture checks while the IT Network Engineering Team maps out the new VLAN architecture. The testing phase follows in weeks three and four. At this stage, engineers will configure a pilot group of 10 VR headsets to connect to the new VLAN through the VPN, verifying posture checks without disrupting normal operations.

Week five marks the deployment phase. Following final approval from Executive Management, the technical teams will push the new network rules and MDM policies to all 150 remote VR devices. Finally, the monitoring phase occurs in week six, allowing the security teams to actively review centralized logs, ensure the policies function as intended, and troubleshoot any remaining connectivity issues.

Resources

Executing this project requires a combination of personnel, technology, and organizational backing. The primary human capital involves dedicated labor hours from both the IT Network Engineering and Information Security teams. Technologically, the initiative relies on the established Cisco routing hardware and AnyConnect VPN gateway, alongside the newly procured MDM Enterprise cloud platform and solid state storage drives. Organizational support includes the budget approved by Executive Management and the communication channels needed to coordinate the rollout with distributed developers.

Dependencies

Several key factors could affect the successful implementation of this solution. A primary internal prerequisite is the timely approval of the project budget by Executive Management, as software licenses cannot be purchased without it. Externally, the project relies on the procurement timeline and cloud availability of the selected MDM vendor. The initiative also assumes the existing Cisco routing hardware has the capacity to handle the new VLAN configurations without requiring unexpected firmware upgrades.

Risk Mitigation

Anticipated risks require proactive mitigation across all four implementation phases. During the initial design, a primary concern is that the proposed MDM policies might conflict with existing VPN rules. The IT and Security teams will prevent this by holding joint architectural review sessions before applying any configurations. As the project moves into the testing phase, the main risk shifts to potentially disrupting active employee training sessions. The team will mitigate this by strictly limiting the pilot to noncritical users and conducting tests outside of core business hours.

Once the VLAN rules go live during deployment, a widespread network connectivity failure becomes the most severe potential risk. A phased rollout approach combined with preconfigured rollback procedures will ensure the team can instantly revert to the previous routing state if major errors occur. Finally, the influx of new audit trails during the monitoring phase could cause the central logging server to run out of storage space. Configuring automated storage alerts and setting up a strict log archiving rotation will effectively prevent this issue.

Benefits and Impact

Security Benefits

The implementation of the proposed technical solution provides several critical security advantages. Primarily, it establishes a Zero Trust architecture that eliminates the risk of lateral network movement. By isolating the VR traffic into a dedicated VLAN, the solution ensures that a compromised headset cannot access the mission critical source code servers. Additionally, enforcing TLS encryption for all transmitted telemetry guarantees the strict confidentiality of employee biometric data.

Operational Impact

This solution also enhances overall performance efficiency and the end user experience. Network performance improves because VR traffic routes through a dedicated subnet rather than congesting the primary corporate infrastructure. Efficiency significantly increases for the IT Network Engineering Team because the MDM platform automates endpoint posture checks, eliminating the need for manual device audits. Finally, the remote developers retain a seamless user experience because background posture checks occur instantly during the standard VPN login process without requiring additional authentication steps.

Business Impact

On a strategic level, the technical solution delivers three major business impacts. It ensures strict regulatory compliance by actively protecting sensitive biometric data from interception, shielding the organization from severe privacy violations and associated legal penalties. It also safeguards the intellectual property of the company by completely isolating the source code servers from vulnerable IoT devices. Ultimately, it creates a highly scalable remote work infrastructure, allowing Executive Management to confidently expand the VR training program without proportionally increasing the corporate attack surface.

Conclusion

In conclusion, the rapid deployment of unmanaged VR headsets created a critical vulnerability that bridges poorly secured home networks directly to the corporate environment. This project proposes a comprehensive Zero Trust solution utilizing an MDM platform, a dedicated VLAN, and strong encryption to secure all remote VR access. Stakeholders should approve this initiative because it actively prevents catastrophic ransomware infections by eliminating lateral movement pathways. Furthermore, it guarantees employee privacy by

encrypting all highly sensitive spatial and biometric telemetry. Ultimately, this implementation is incredibly cost-effective and efficient because it seamlessly integrates with the existing Cisco networking infrastructure.

References

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Appendix A

A Proposed Network Topology Diagram:

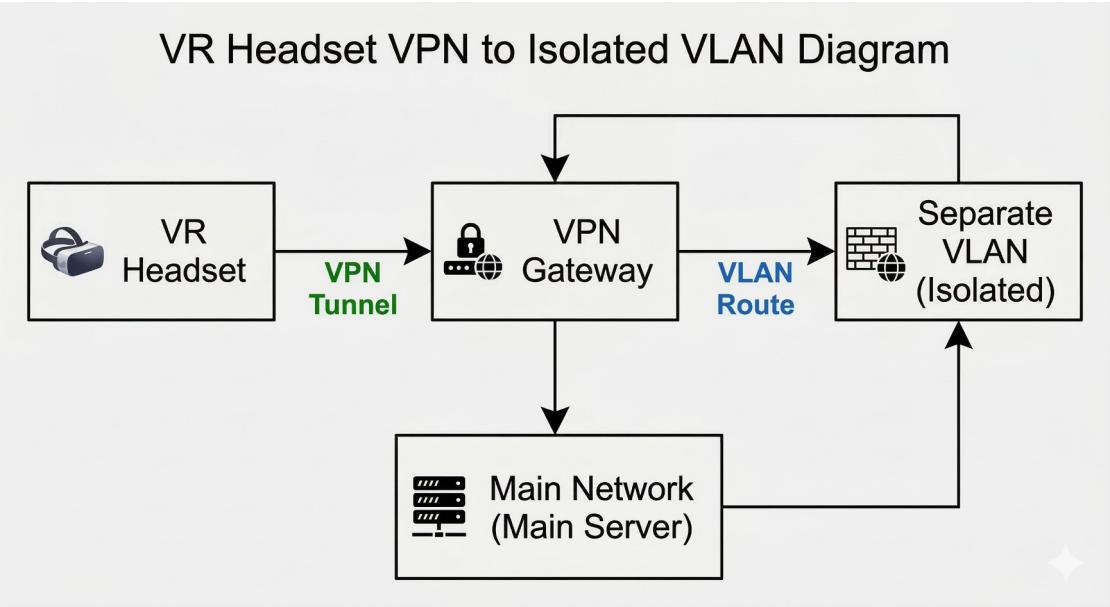


Figure 1: Picture of original network topology diagram created by the author.

Appendix B

Cloud MDM Vendor Quote:

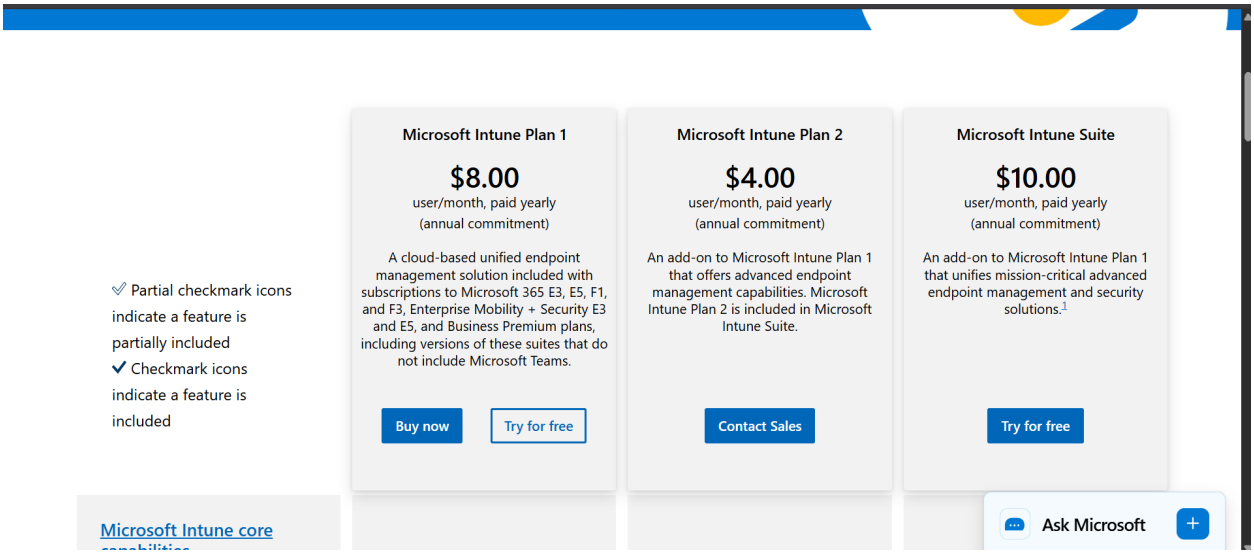
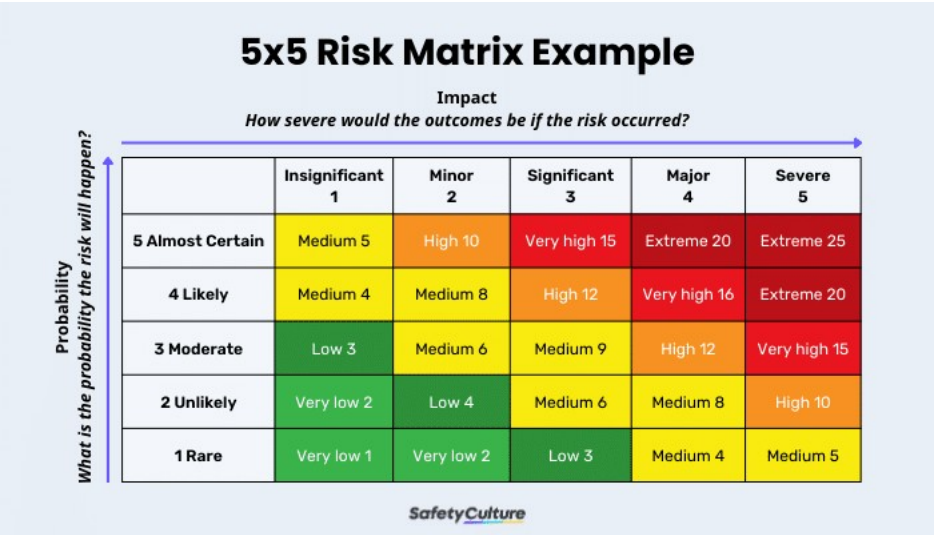


Figure 2: Adapted from Microsoft Intune pricing, by Microsoft, n.d., Retrieved March 10, 2026, from https://www.microsoft.com/en-us/security/business/microsoft-intune-pricing

Appendix C

Detailed Risk Assessment Matrix:



5x5 Risk Matrix Example

Impact

How severe would the outcomes be if the risk occurred?

Probability

What is the probability the risk will happen?

	Insignificant 1	Minor 2	Significant 3	Major 4	Severe 5
5 Almost Certain	Medium 5	High 10	Very high 15	Extreme 20	Extreme 25
4 Likely	Medium 4	Medium 8	High 12	Very high 16	Extreme 20
3 Moderate	Low 3	Medium 6	Medium 9	High 12	Very high 15
2 Unlikely	Very low 2	Low 4	Medium 6	Medium 8	High 10
1 Rare	Very low 1	Very low 2	Low 3	Medium 4	Medium 5

SafetyCulture

Figure 3: Adapted from 5x5 risk matrix, by SafetyCulture, n.d., Retrieved March 10, 2026, from https://safetyculture.com/topics/risk-assessment/5x5-risk-matrix